Zhenlin (Cassie) Zhu

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SUMMARY

PhD Data Scientist with **6+** years of experience developing statistical models, deep learning systems, and end-to-end data and machine learning pipelines, including ETL, feature engineering, model training, evaluation and dashboarding across cloud environments. Strong foundation in probability, hypothesis testing, simulation, and optimization for decision support. Experienced in predictive modeling, A/B testing, GenAI/LLM, and large-scale data processing. Effective communicator skilled at translating ambiguous stakeholder needs into well-formulated analytical problems and delivering actionable insights to technical and non-technical teams.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL
ML/AI: Deep Learning, LLMs/GenAI, RAG, Time-series Forecasting, Gradient Boosting, Random Forests, Fraud Detection
Data & MLOps: Feature Engineering, A/B Testing, AWS, PySpark, MLflow, Model Deployment, Model Monitoring
Visualization: Tableau, PowerBI, Matplotlib, Seaborn

WORKING EXPERIENCE

Machine Learning Engineer Intern | *Eth Tech* Newark, CA | May 2025 – Aug 2025

- Fine-tuned Salesforce's **BLIP** imaging captioning model, **improving BLEU-4 scores by 16%** over baseline.
- Built LLM-powered product metadata enrichment with **Sentence-Transformer** and **name entity recognition**.
- Developed a cohort retrieval system based on product click-through rates and automated updates with Airflow, improving recommendation freshness.
- Built a **two-tower retrieval model** with negative sampling; boosted **CTR by 1.2%** in **A/B testing**.
- Implemented **incremental training** workflows with **Airflow** for automated model refresh, improving total user actions by **0.6%**.

Research Scientist Postdoctoral Scholar | *UCLA* Los Angeles, CA | Nov 2024 – Present

- Built an end-to-end **feature extraction** and multi-class classification pipeline from raw data, training tree-based models (Random Forest, XGBoost, LightGBM) on **5,000+ samples** and achieving **0.93 macro ROC-AUC.**
- Achieved robust **multi-class classification** across four categories using minimal input features, with interpretable feature importance supporting model validation and downstream analysis.
- Collaborated with cross-disciplinary teams, translating broad questions into structured analytical problems and communicating outcomes to diverse audiences.

PhD Researcher | *Netherlands Institute for Space Research* Leiden, Netherlands | Sep 2019 – Oct 2024

- Developed **scalable SQL and Python pipelines** to collect, process and visualize **large-scale unstructured astronomical data**, revealing hidden physical structures using adaptive-binning and image-sharpening techniques.
- Implemented **statistical anomaly detection**, using **Fisher-Pearson skewness coefficient** to quantify flux distribution asymmetry and detect statistically significant outliers as gas clumps candidates.

Data Scientist Intern | *Pivigo* London, UK | Jun 2022 – Sep 2022

- Built **a recommendation system** for retailer platforms by processing **100k+ products and ~1M** anonymized customer records, combining **CNNs, GRU4Rec and Word2Vec** for hybrid approaches.
- Optimized ranking performance using learning-to-rank objectives in **LightGBM and CatBoost** (e.g., LambdaRank / YetiRank), improving **mean average precision by 2.4×** compared to a heuristic ranking baseline.
- Delivered actionable insights to product teams, improving recommendation relevance and supporting data-driven personalization strategies.

SELECTED PROJECTS

Paper QA Chatbot using RAG: Built an end-to-end GenAI **Retrieval-Augmented Generation (RAG)** chatbot to assist with academic paper comprehension. Implemented PDF parsing, text chunking and embedding generation with OpenAI API, storing semantic indexing in **FAISS** for fast retrieval. Integrated **LangChain's RetrievalQA** chain with GPT-4 to improve the accuracy of reponses to user queries. Deployed a responsive **Streamlit app** that supported dynamic paper uploads and interactive querying.

Predicting Potential Customers: Designed and implemented **a decision tree model** to identify high-conversion leads using behavioral and demographic features. Improved prediction accuracy **from 84% to 90%** by applying **GridSearchCV hyperparameter tuning**, reducing overfitting. Explored random forest modeling to analyze feature importance, identifying top predictors including time on site, entry channel, pages per visit, and age.

EDUCATION

Ph.D in Astrophysics *Leiden University* Sep 2019 –Oct 2024
B.S & M.S. in Astrophysics *Nanjing University* Sep 2012 – Jul 2019
- 15 peer-reviewed papers (6 as first author), 30+ talks at leading conferences around the world.